MIL-G-80126B
27 February 1987
SUPERSEDING
MIL-G-80126A
4 December 1972

MILITARY SPECIFICATION

GRINDING MACHINES, DISC, HORIZONTAL, DOUBLE END, SINGLE SPINDLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of floor mounted, electric motor driven, horizontal, double end, single spindle disc grinding machines with equipment and accessories specified herein.

1.2 Classification. The machine shall be of the following sizes. The size to be furnished shall be as specified (see 6.2.1).

Size	Disc diameter
12	12-inch
14	14-inch
18	18-inch
20	20-inch
24	24-inch
26	26-inch
30	30-inch
36	36-inch
42	42-inch

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE REFERENCE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification as specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U.S. DEPARTMENT OF LABOR

29 CFR 1910 - Occupational Safety and Health Standards (OSHA).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402-0001).

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified. Unless otherwise specified, the issue of the DoD adopted documents shall be as listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issue of documents not listed in the DoDISS shall be the issue of the non-Government documents which are current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NEMA MG 1	- Motors and Generators.
ANSI/NFPA 70	- National Electrical Code.
ANSI/NFPA 79	- Electrical Standard for Industrial Machinery.
ANSI B5.1	- T-Slots, Their Bolts, Nuts and Tongues.
ANSI B5.35	- Machine Mounting Specifications for Abrasive Discs and Plate Mounted Wheels.
ANSI B7.1	- American Standard Safety Code for the Use, Care, and Protection of Abrasive Wheels.
ANSI B11.9	- Safety Requirements for the Construction, Care, and Use of Grinding Machines.
ANSI B74.2	- Shapes and Sizes of Grinding Wheels.
ANSI B74.13	- Marking for Identifying Grinding Wheels and Other Bonded Abrasives.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3351.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951	- Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103-1187).

AMERICAN GEAR MANUFACTURER'S ASSOCIATION (AGMA)

AGMA 360.02	- Manual for Machine Tool Gearing.
AGMA 390.03	- Gear Classification, Materials and Measuring Methods For Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturer's Association, Standards Department, Suite 1000, 1901 North Fort Meyer Drive, Arlington, VA 22209-1695.)

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INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 54 - Cylindrical Gears For General Engineering and For Heavy Engineering, Modules and Diametral Pitches of.

(Application for copies should be addressed to the International Organization for Standardization, c/o American National Standards Institute, ATTN: Sales Department, 1430 Broadway, New York, NY 10018-3351).

(Non-Government standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. If the text of this specification conflicts with the references cited (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The machine shall be new and one of the manufacturer's current models. Each machine shall grind flat, smooth surfaces on castings, forgings, bar stock, workparts, and various tools in accordance with the performance requirements specified herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as scales, depth stops, carriage stops, dial indicators,

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pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials if those materials will not jeopardize the machine's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled to restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency as governed by the developments available within the industry.

3.2.4 Controls. All operating controls shall be located convenient to the operator at his normal work station.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of ANSI B11.9 and OSHA 29 CFR 1910 that are applicable to the machine itself. If a conflict occurs between the requirements of ANSI B11.9 and OSHA 29 CFR 1910, the requirements of OSHA 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.4).

3.2.6 Mercury restriction. The machine shall not contain mercury nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. Under operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prescribed by Federal, state, or local statutes in effect at the point of installation (see 6.4).

3.2.9 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. Recirculating lubrication systems shall include

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a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.10 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed to prevent loosening. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the machine and related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Dials. All rotating dials or handwheels which are used to position machine axes or for tool movement shall be graduated in increments not larger than 0.001 inch. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduation that can be read from the operator's normal position.

3.3.8 Gears. All gears and pinions of spindle and axis drive trains shall be designed and manufactured of a suitable ferrous material to meet or exceed the requirements of AGMA 360.02 and 390.03 for the English (US)

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system or ISO 54 for the metric (SI) system. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in the drive train shall be hardened and ground steel. Hardness shall be not less than Rockwell C-45.

3.3.9 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirements shall be as specified.

3.4 Components. Each machine shall consist of not less than a base, two tables, spindle assembly with abrasive disc wheels, driving mechanism, and operating controls. Any additional components necessary for the operation of the machine shall be furnished in accordance with the manufacturer's standard practice.

3.4.1 Base. The base shall be a rectangular, double-walled, box-type casting or a stress-relieved steel weldment, ribbed and braced. The spindle assembly and abrasive disc wheels shall be supported on the base as a single unit. Means for securing the machine to the floor shall be provided.

3.4.2 Tables. Tables shall be plain or lever type, left- or right-hand mounted. Unless otherwise specified (see 6.2.1), each machine shall be provided with one left-hand plain table and one right-hand lever feed table, mounted to the respective end of the spindle, perpendicular to the face of the abrasive disc wheel. Each table shall be horizontally and vertically adjustable for proper positioning of the table relative to the disc face and spindle centerline. Tables shall tilt for angular grinding. Angular adjustment shall be indicated on a protractor in degrees of tilt. Means of locking the tables after positioning shall be provided. Each table shall have a machined working surface.

3.4.2.1 Plain table. The plain table shall be used for free hand grinding of flat surfaces. Workpieces shall rest on the table top and pressed manually against the abrasive disc wheel.

3.4.2.2 Lever table. The lever table shall be used for large, awkward workpieces or workpieces requiring heavy cuts. The operator shall pull up on the lever handle to feed workpieces into the abrasive disc wheel. Table in-feed shall be regulated by a graduated, adjustable micrometer stop for accurate stock removal and size duplication. Unless otherwise specified (see 6.2.1), table in-feed and oscillation shall be manually controlled. Two or more T-slots conforming to ANSI B5.1 shall be provided the full length of the table surface for mounting workpieces with fixtures. Each lever table shall be equipped with an adjustable counterbalance.

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3.4.3 Spindle assembly. The spindle assembly shall include a work spindle and related components, spindle housing, drive mechanism, and abrasive disc wheels. The spindle shall be constructed of alloy steel, machine finished, horozontally mounted on two or more preloaded ball or roller bearings designed to minimize radial and axial play, and shall be dynamically balanced. The spindle shall accept an abrasive disc wheel on each end. Spindle drive shall be either direct motor drive with the motor shaft serving as the work spindle or by belts and pulley. Spindle and motor bearings shall be protected from dust and contaminants by seals. Means shall be provided to lubricate the spindle and bearings (except the sealed-for-life type). Spindle mounting shall compensate for spindle expansion and contraction due to temperature fluctuation.

3.4.4 Abrasive disc wheels. Disc wheels shall be constructed of carbon steel with a machined face surface. One disc wheel shall be mounted on each end of the spindle and shall rotate with the spindle. Disc wheels shall be dynamically balanced. Means shall be provided to mount an abrasive disc to each disc wheel face in accordance with the requirements given in ANSI B5.35. Unless otherwise specified (see 6.2.1), the abrasive discs shall conform to ANSI B74.2, be marked in accordance with ANSI B74.13, and meet the safety requirements established in ANSI B7.1. A steel backing plate or wheel collar shall be provided with each abrasive disc wheel to support the wheel and assure rigidity during grinding. Guards or protective shields shall be provided on each abrasive disc wheel.

3.4.5 Wheel dresser. Two manually operated wheel dressers shall be provided with each machine. One dresser shall be mounted on each table. Each wheel dresser shall dress the face of each abrasive disc wheel. Dressers shall be adjustable to feed into the abrasive disc wheels. Dressing shall be accomplished by manually pivoting the table-mounted dressers past the abrasive disc wheels.

3.4.6 Coolant system. When specified (see 6.2.1), a coolant system shall be provided. The coolant system shall include a sump or reservoir, a power driven pump and all necessary piping. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining and cleaning and shall include a baffle or a strainer which is easily removable for cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant.

3.4.7 Dust collector. When specified (see 6.2.1), a dust collection unit shall be provided. The dust collector shall be connected to the machine's electrical system with start/stop pushbuttons conveniently located on the control panel. The dust collector shall be either a built-in unit or a separate external unit.

3.4.8 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to ANSI/NFPA 79. Each machine shall draw all of its electrical power from a single 230/460-volt, 3-phase, 60-hertz (Hz) circuit. Each machine shall have a fused safety disconnect switch or

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circuit breaker. The machine shall be initially wired for operation on 460 volts. An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in ANSI/NFPA 79 for the disconnect fuse rating.

3.4.8.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Alternating-current (AC) motors shall be designed to operate on 60-Hz. Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements for a dripproof enclosure. All motors shall meet the requirements of ANSI/NEMA MG-1 and ANSI/NFPA-70.

3.4.9 Hydraulic system. The hydraulic system, when provided, shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level, a provision for draining, and a cleanout access plate if the reservoir is not removable.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the machine shall be not less than specified in table I for the machine size specified.

3.6 Performance. Each machine shall be capable of grinding surfaces flat to within 0.004 inch per 4 inches and square to within 0.005 inch per 4 inches. Performance shall be tested in accordance with the procedures described in 4.7.1 and 4.7.3.

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.7 Alignment tolerances. Each machine shall meet the alignment tolerances specified in table II when tested in accordance with 4.7.2.

3.8 Standard equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished with each machine:

a. One set of special hand tools normally furnished by the manufacturer.

b. Abrasive disc, number, type, dimensional characteristics, and composition to be as specified (see 6.2.1).

c. Two disc wheel safety hoods with rear exhaust outlets.

d. Two plain eye shields.

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3.9 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2.1).

3.10 Repair parts. Specified repair parts (see 6.2.1) shall be furnished.

3.11 Marking on plates. All words on plates shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background. All instruction and identification plates shall be of corrosion-resistant material.

3.11.1 Lubrication plate. Unless otherwise specified (see 6.2.1), a lubrication plate shall be permanently and securely attached to each machine. The plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.11.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2.1), the nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.12 Hour meter. When specified (see 6.2.1), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of not more than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.13 Technical data. When technical data (operating manuals, maintenance manuals, parts catalogs, prints, wiring diagrams, lubrication charts, and machine alignment and accuracy test results) is required, it shall be furnished in accordance with the requirements of DD Form 1423. All technical data furnished shall be written in the English language.

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3.14 Workmanship. Workmanship of the machine shall be commensurate with all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2.1), all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under the normal environmental conditions.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6, all tests in 4.7, and the inspection in 4.8. Failure of the item to pass the first article examination or any tests shall be cause for rejection.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.6, the test in 4.7.1, and the

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inspection in 4.8. Failure of the item to pass the examination, the tests, or the inspections shall be cause for rejection.

4.6 Examination. The machine shall be visually examined to determine compliance with requirements of this specification as given in 3.2 through 3.5 and 3.8 through 3.14, including all of the referenced documents.

4.7 Tests.

4.7.1 Operational test. The machine shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.2 Alignment accuracy test. Each machine shall be tested to assure compliance with the alignment accuracy requirements given in table II. The instruments and methods used to perform the tests shall be at the supplier's option provided they accurately determine the tolerances of the machine.

4.7.3 Performance tests. Each machine shall be tested using a test piece of SAE 1040 steel, approximately 2 inches thick, 3 inches wide, and 5 inches long. Abrasive wheel grade and grain size to be used during the test shall be manufacturer's option. Grind at least 0.010 inch of stock from one end and one side of the test piece. At the option of the procuring agency, each machine shall be subjected to tests in accordance with specific requirements designated by the procuring activity in lieu of or in addition to the above test (see 6.2.1).

4.7.4 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.9.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The grinding machines covered by this specification are for free hand finishing of flat, smooth surfaces on workpieces, bar stock, castings, forgings, and various tools as required in general machine shops.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Size required (see 1.2).

c. First article, if required (see 3.1).

d. If machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Additional safety and health requirements, if required (see 3.2.5).

g. Painting, if different (see 3.3.5).

h. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.9).

i. Tables, if different (see 3.4.2).

j. Lever table operation, if different (see 3.4.2.2).

k. Abrasive disc wheels, if different (see 3.4.4).

l. Coolant system, if required (see 3.4.6).

m. Dust collector, if required (see 3.4.7).

n. Electrical system and input power, if different (see 3.4.8).

o. Motor enclosure, if different (see 3.4.8.1).

p. Size and capacity, if different (see 3.5).

q. Additional performance features, if required (see 3.6.1).

r. Standard equipment, if different (see 3.8).

s. Abrasive disc number, type, dimensional characteristics, and composition, as specified (see 3.8(b)).

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t. Specify and fully describe optional equipment, if required (see 3.9).

u. Specify applicable repair parts, if required (see 3.10).

v. Lubrication plate, if different (see 3.11.1).

w. Nameplate, if different (see 3.11.2).

x. Hour meter, if required (see 3.12).

y. Inspection conditions, if different (see 4.3).

z. First article inspection, if different (see 4.4).

aa. Quality conformance inspection, if different (see 4.5).

bb. Performance test, if different (see 4.7.3).

cc. If preservation, packing, and marking in accordance with MIL-M-18058 are required, specify level required and specification number (see 5.1).

6.2.2 Contract data requirements. Required technical data (operating manuals, parts lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports) should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern "the machine in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat. Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual machine in the environment), the procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional requirement that will integrate the new machine into its future operating environment.

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6.5 Cross-reference of classification changes. Machine classification changes from the previous issue of this specification are as follows:

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Size 8	Deleted
Size 10	Deleted
Size 12	Size 12
Size 14	Size 14
Size 18	Size 18
Size 20	Size 20
Size 23	Deleted
Size 24	Size 24
Size 26	Size 26
Size 30	Size 30
Size 36	Size 36
Size 42	Size 42

6.6 Subject term (key word) listing.

Abrasive wheel, grinding
Disc, double end
Grinder, disc
Grinding machine, disc
Machine tool, grinding
Spindle, horizontal, single
Wheel, abrasive, grinding

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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TABLE I. Sizes and capacities.

Characteristics	Requirements								
Capacity:	Size 12	Size 14	Size 18	Size 20	Size 24	Size 26	Size 30	Size 36	Size 42
Steel disc wheel, diameter, inches	12	14	18	20	24	26	30	36	42
Abrasive disc wheel, diameter, inches	12	14	18	20	24	26	30	36	42
Table work surface, inches									
Plain	5-1/2 x 10	5-1/2 x 12	6-1/2 x 13	< ----- 8	x 14 ------ >	10 x 16	14 x 20	12 x 30	12 x 30
Lever or Universal	5 x 9-5/8	5 x 9-5/8	6 x 11	< --- 7	x 13-1/2 ---- >	<------	------- 8	x 14 -------	--------->
Table in-feed adjustment, inches	2	2	< ----	----------	--- 3 --------	----------	-------- >	4	4
Table tilt adjustment, degrees									
Up	< ------	----------	----------	----------	----- 20 -----	----------	----------	----------	-------- >
Down	< ------	----------	----------	----------	----- 45 -----	----------	----------	----------	-------- >
Spindle centerline to floor, maximum, inches	< ------	----------	----------	----------	----- 42 -----	----------	----------	----------	-------- >
+Wheel collars, diameter, inches	4	5	6	< -----	----- 12 -----	--- >	14	17	17
Speeds:									
Spindle, RPM	1800	1800	1200	1200	900	900	900	900	900
Motor horsepower:									
*Spindle	2	3/5	3/5/7-1/2	7-1/2/10	10/15	15/20/25	15/20/25	20/25	20/25/30

* Optional spindle motor horsepower available as indicated.

\+ When applicable.

NOTE: Machine sizes and capacities shall be not less than the stated requirements of the size ordered nor greater than the requirements of the next larger size shown in above table. When the largest size and capacity machine shown in above table is ordered, the size and capacity of the machine offered shall not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater.

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Table II. Alignment accuracy requirements.

Test to be applied	Permissable deviation
Spindle lateral thrust movement	0.005" TIR
Spindle and steel disc wheel radial runout, hand revolving, slow speed	0.003" TIR
Tables square with steel disc wheel	±0.003"
Steel disc wheel face runout Size 12 Size 14 and above	 ±0.003" TIR ±0.005" TIR
Table oscillation parallel to wheel face (when applicable)	±0.005" Total

Custodians:
Air Force - 99

Preparing activity:
DLA - IP

Project (3415-0160)

Review activities:
Air Force - 84
DLA - GS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-80126B	Grinding Machines, Disc, Horizontal, Double End, Single Spindle

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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